In[ ]:=
```
n1 = 10;
g1 = 3;
n2 = 5;
g2 = 2;
n3 = 5;
g3 = 2;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.5068195691852881;
β12 = 0.20660077501801427;
β22bar = 0.7778524931404934;

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
 aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2) / 5;
theta = -6 η / (5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.5326530612244897; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -1.3456132586363403`;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
```

Q3 $\frac{f3^{\frac{3}{2}}}{\sqrt{x}}$ in the middle region. This is obtained by solving two

consistency conditions along with other constant γ12bar. *)

```
Q2 = -3.6465672354338867;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
```

Q2 $\frac{f3^{\frac{3}{2}}}{\sqrt{x}}$ in the inner region. This is obtained by solving three

consistency conditions along with other constants β13 & β23. *)

```
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
```

$$R = \frac{3\,a}{(8 * alfas)} * \left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{2}{3}};$$(*Page number 54, it is inverse in the copy though *)

$$c = \frac{12}{5\,\pi}\,\frac{4}{3}\,g^2 - \frac{4}{3}\,\frac{alfas}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}};$$

(*Multiplication factor in the expression of kinetic energy *)

$$d = \frac{4}{3}\,g^2 - \frac{4}{3}\,\frac{alfas}{a}\left(\frac{3\,\pi\,\eta}{2}\right)^{\frac{1}{3}}\,\frac{18}{25\,\pi}\,\frac{\eta}{\left(\eta - \frac{1}{5}\right)};\quad (* \;\frac{18g^2}{5(5\eta\,-1\,)}.16\;\pi^2.\left(\frac{3a}{8alfas}\right).\;\frac{R^2}{(6\pi^2)^2}\;,$$

The value of R is from page number 54 which is written inverse here,

and $\left(\frac{3a}{8alfas}\right)$ is inverse in copy, BA, 13 September *)

(*Multiplication factor in the expression of Potential energy *)

(*Let us define the three functions for three regions *)

```
f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
  Abs[x3 * y3'[x3] - y3[x3] - n3 * Q3/(3 η) - β22bar * n2 * Q2/(3 η) + β22bar^(3/2) * n2 * Q3/(3 η)]
   (*Appendix from Kinetic energy*) /.
```

```
First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)), y3'[x2] == β22bar * s2 ,
    y3[x2] == β22bar * u2 (* It shows how the function f3
     (x3) changes on other regions*)}, y3, {x, x2, x3}];
(* s2 is the first derivative of function f_2(x) *)
(*u2 is the function f_2(x) *)
(*This is the function f3 defined on the
  middle region. It has the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
    γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
    γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
   (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
     y1'[x3] == γ12bar * s3, (* s3 is the first derivative of function f_2bar(x) *)
     (* It means at the boundary x3 the function df1 changes to other function *)
     y1[x3] == γ12bar * u3 (* u3 is the function f_2bar(x) *)
     (*It means at the boundary x3 the function f1 changes to other function*)
    }, (* It means at the boundary x3 the function f1 changes to other function *)
    y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
 Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
   condition of (d/dx) f1(x)|x2 = γ12bar (d/dx) f2(x)|x2 and f1(x3)=γ12 f2(x3)  *)
f2[xmax_?NumericQ, m2_?NumericQ] :=
```

```
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^(3/2)] / x^(1/2)), y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];
   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3 / 2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2 / (3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x_3 inner region. On this region,
f_1(x)=β_13 f_3(x) and f_2(x)= β_23 f_3(x).   *)

(*Start the program*)

m2 = 45;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
  m2B = m2;
  Energy = 10^9;
  Energy = 10^9;
  Energy2plus = 10^9;
  H = 10^9 + 1;
  SN = 0;
```

```
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]]^3/2) / x^1/2,
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]]; (* u2 is the function f_2(x) *)
  s2 = t4[[1]]; (* s2 is the first derivative of function f_2(x) *)
  If[u2 < 0., u2 = 0.];


  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2,  u2,  s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2,  u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]]^3/2) / x^1/2,
     y3'[x2] == β22bar* s2 , y3[x2] == β22bar* u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]]; (* u3 is the function f_2bar(x) *)
  s3 = t5[[1]]; (* s3 is the first derivative of function f_2bar(x) *)
  If[u3 < 0., u3 = 0.];

  x1 = x3 + 0.1;
  FindMinimum[{f1[x1, x3,  s3, u3]}, {x1, x3}];
  x1 = x1 /. Last[FindMinimum[{f1[x1, x3,  s3, u3]}, {x1, x3}]];
```

$$\text{solution1} = \text{NDSolve}\left[\left\{y1''[x] == \left(Q1 * \frac{\text{Abs}\left[y1[x]^{\frac{3}{2}}\right]}{x^{\frac{1}{2}}}\right),\right.\right.$$

$$\left.\left. y1'[x3] == \gamma 12\text{bar} * s3 \, , \, y1[x3] == \gamma 12\text{bar} * u3 \right\}, y1, \{x, x3, x1\}\right];$$

```
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]]; (* u1 is the function f_1(x) *)
s1 = t6[[1]]; (* s1 is the first derivative of function f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)
```

$$\text{normmx2} = x2\, s2 - u2 ;$$

$$\text{normx2} = \frac{n2\, Q2}{3\, \eta} ;$$

```
(*normx2barr and normx2bar should be equal *)
```

$$\text{normmx3} = x3\, s3 - u3 ;$$

$$\text{normx3} = \frac{n3\, Q3}{3\, \eta} + \beta 22\text{bar} \frac{n2\, Q2}{3\, \eta} - (\beta 22\text{bar})^{\frac{3}{2}} \frac{n2\, Q3}{3\, \eta};$$

```
(*normmx2 and normx2 should be equal *)
```

$$\text{normmx1} = x1\, s1 - u1 ;$$

$$\text{normx1} = \frac{n1\, Q1}{3\, \eta} + \gamma 12\text{bar} \frac{n3\, Q3}{3\, \eta} - (\gamma 12\text{bar})^{\frac{3}{2}} \frac{n3\, Q1}{3\, \eta} +$$

$$\left((\gamma 12\text{bar})^{\frac{3}{2}} (\beta 22\text{bar})^{\frac{3}{2}} - (\beta 12)^{\frac{3}{2}}\right) \frac{n2\, Q1}{3\, \eta} - \gamma 12\text{bar} (\beta 22\text{bar})^{\frac{3}{2}} \frac{n2\, Q3}{3\, \eta} + \gamma 12\text{bar}\, \beta 22\text{bar} \frac{n2\, Q2}{3\, \eta} ;$$

```
(*normmx1 and normx1 should be equal *)


h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
```

```
h6 = h3[[1]] ;
AB = 4/7 Sqrt[x2] (u2)^(5/2) - 5 n2/(21 η) s2 ;
CD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 (β22bar)^(5/2) Sqrt[x2] (u2)^(5/2) -
   5/(7 Q3) s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) + 5/(7 Q3) (β22bar)^2 n2 Q2/(3 η) s2 ;
CCD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 Sqrt[x2] (β22bar)^(5/2) (u2)^(5/2) - 5/(7 Q3) s3 normmx3 +
   5/(7 Q3) (β22bar)^2 normmx2 s2 ;

EF = 4/7 Sqrt[x1] (u1)^(5/2) - 4/7 (γ12bar)^(5/2) Sqrt[x3] (u3)^(5/2) -
   5/(7 Q1) s1 (n1 Q1/(3 η) + γ12bar n3 Q3/(3 η) - (γ12bar)^(3/2) n3 Q1/(3 η) + n2 Q1/(3 η) ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) -
      (γ12bar) (β22bar)^(3/2) n2 Q3/(3 η) + (γ12bar) (β22bar) n2 Q2/(3 η)) +
   5/(7 Q1) (γ12bar)^2 s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) ;
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)

K222 = n2/(3 η Q2) s2 - 1/Q2 h4 ;
K22 = 4 n2/(7 η Q2) s2 - 4/(7 Q2) Sqrt[x2] (u2)^(5/2) ;
(*K2bar2barr and K2bar2bar should be equal *)
K32 = - 4/(7 Q2) β22bar^1.5 Sqrt[x2] (u2)^(5/2) +
```

$$\frac{n2}{3\,\eta}\left(\frac{\beta22bar^{1.5}}{Q2}-\frac{\beta22bar}{Q3}+\frac{5*\beta22bar^{1.5}}{7\,Q2}\right)s2+\frac{n2}{3\,\eta\,Q3}s3\,;$$

$$K332=\frac{n2\,(\beta22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}s2-\frac{n2\,\beta22bar}{3\,\eta\,Q3}s2+\frac{n2}{3\,\eta\,Q3}s3-\frac{(\beta22bar)^{\frac{3}{2}}}{Q2}h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21=-\frac{4}{7\,Q2}\beta12^{1.5}\sqrt{x2}\;(u2)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\beta12^{1.5}+\frac{\beta12^{1.5}}{Q2}-\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}\right)s2+$$

$$\frac{n2}{3\,\eta}\left(\frac{\gamma12bar^{1.5}}{Q3}-\frac{\gamma12bar}{Q1}\right)s3+\frac{n2}{3\,\eta\,Q1}s1\,;$$

$$K221=\frac{n2}{3\,\eta\,Q2}(\beta12)^{\frac{3}{2}}s2+\frac{n2\,(\gamma12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}s3-$$

$$\frac{n2}{3\,\eta\,Q3}(\gamma12bar)^{\frac{3}{2}}\beta22bar\,s2+\frac{n2}{3\,\eta\,Q1}s1-\frac{n2}{3\,\eta\,Q1}\gamma12bar\,s3-\frac{(\beta12)^{\frac{3}{2}}}{Q2}h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31=\frac{4}{7}\beta22bar^{1.5}\left(\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}-\frac{\beta12^{1.5}}{Q2}\right)\sqrt{x2}\;(u2)^{\frac{5}{2}}-\frac{4}{7\,Q3}\gamma12bar^{1.5}\sqrt{x3}\;(u3)^{\frac{5}{2}}+$$

$$\frac{n2}{3\,\eta}\beta22bar\left(\frac{5}{7\,Q2}\beta22bar^{\frac{1}{2}}*\beta12^{1.5}-\frac{\gamma12bar^{1.5}*\beta22bar^{1.5}}{Q3}-\frac{5\,Q2}{7\,Q3^2}\gamma12bar^{1.5}*\right.$$

$$\left.\beta22bar+\frac{Q2}{Q3^2}\gamma12bar^{1.5}*\beta22bar+\frac{1}{Q2}\beta12^{1.5}*\beta22bar^{\frac{1}{2}}-\frac{\beta12^{1.5}}{Q3}\right)s2+$$

$$\frac{1}{3\,\eta\,Q3}\left(n3*\gamma12bar^{1.5}-\frac{n3\,Q3}{Q1}\gamma12bar+n2*\beta12^{1.5}-\frac{n2\,Q2}{Q3}\gamma12bar^{1.5}*\beta22bar+\right.$$

$$\left.\frac{5}{7}\gamma12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\beta22bar-n2*\beta22bar^{1.5}\right)\right)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

$$(*K122 = \frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$- \frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] + \frac{n3}{3\,\eta\,Q1}s1 - \frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+ (\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\left[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2 + \frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] +$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\ ;\ *)$$

$$K331 = \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2 - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5 -$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2 + \frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2} - \frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2 + \frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3} - \frac{1}{Q1}\right)s3 + \frac{n3}{3\,\eta\,Q1}s1\ ;$$

(*K12 and K122 should be equal*)

$$K33 = \frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3} - \frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2} + \frac{2}{7}\frac{Q2}{Q3^2} - \frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2 +$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3} + \beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}} - \frac{2}{7}\frac{Q2}{Q3} - \frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\ ;$$

$$K333 = \frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s3 - \frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s2 -$$

$$\frac{(\beta 22bar)^3}{Q2}h4 - \frac{1}{Q3}h5 + (\beta 22bar)^3\frac{n2}{3\,\eta\,Q2}s2 - (\beta 22bar)^{\frac{5}{2}}\frac{n2}{3\,\eta\,Q3}s2 + \frac{n3}{3\,\eta\,Q3}s3\ ;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta22bar^{\frac{5}{2}} * \gamma12bar^3}{Q3} - \frac{\beta12^3}{Q2}\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma12bar^3}{Q3}\right)\sqrt{x3}\ (u3)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q1}\sqrt{x1}\ (u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{\beta12^{\frac{3}{2}} * \gamma12bar^{\frac{3}{2}} * \beta22bar}{Q3} + \frac{Q2}{Q3^2}\gamma12bar^3 *\right.\right.$$

$$\left.\beta22bar^2 - \frac{\beta12^{1.5} * \gamma12bar^{1.5} * \beta22bar}{Q3}\right) +$$

$$\left.\frac{5\,n2}{21\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{Q2}{Q3^2}\beta22bar^2 * \gamma12bar^3\right)\right)s2 +$$

$$\left(\frac{n2}{3\,\eta}\left(\frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{\beta12^{1.5} * \gamma12bar}{Q1} - \frac{\beta22bar^{1.5} * \gamma12bar^3}{Q3} +\right.\right.$$

$$\frac{\beta22bar^{1.5} * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar -$$

$$\frac{\beta12^{1.5} * \gamma12bar}{Q1} + \frac{\gamma12bar^{\frac{5}{2}} * \beta22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\gamma12bar^2 * \beta22bar -$$

$$\left.\frac{Q3}{Q1^2}\beta22bar^{1.5} * \gamma12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma12bar^3}{Q3} - \frac{2 * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\gamma12bar^2\right) +$$

$$\frac{5}{21\,\eta}\left(\frac{n3}{Q3}\gamma12bar^3 + \frac{n2\ Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{n2}{Q3}\beta22bar^{1.5} * \gamma12bar^3 -\right.$$

$$\left.\left.\frac{n3\ Q3}{Q1^2}\gamma12bar^2 - \frac{n2\ Q2}{Q1^2}\gamma12bar^2 * \beta22bar + \frac{n2\ Q3}{Q1^2}\gamma12bar^2 * \beta22bar^{1.5}\right)\right) *$$

$$s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma12bar + \frac{n3}{3\,\eta\,Q1}\gamma12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}\right.$$

$$\left(\beta 12^{1.5} - \gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{Q2}{Q1}\beta 22bar * \gamma 12bar + \frac{Q3}{Q1}\beta 22bar^{1.5} * \gamma 12bar\right) + \frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\gamma 12bar - \frac{n3}{Q1}\gamma 12bar^{1.5} + \frac{n2}{Q1}\gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{n2}{Q1}\beta 12^{1.5} - \frac{n2\ Q3}{Q1^2}\beta 22bar^{1.5} * \gamma 12bar + \frac{n2\ Q2}{Q1^2}\gamma 12bar\ \beta 22bar\right)\Bigg) * s1;$$

$$K111 = \frac{-(\beta 12)^3}{Q2}h4 - \frac{(\gamma 12bar)^3}{Q3}h5 - \frac{1}{Q1}h6 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3\,\beta 22bar\,s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3(\beta 22bar)^2 s2 + \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma 12bar - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\beta 22bar\,\gamma 12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}\gamma 12bar\right)s1 - \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}(\gamma 12bar)^2 - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}(\gamma 12bar)^2\,\beta 22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^2\right)s3 + \frac{n2}{3\,\eta}\frac{(\beta 12)^3}{Q2}s2 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar\,s3 + \frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^3 s3 - \frac{n2}{3\,\eta\,Q3}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^3 s3 - \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n2}{3\,\eta\,Q1}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n1}{3\,\eta\,Q1}s1;$$

(*K11 and K111 should be equal *)

$$T1 = c\left(\frac{4}{7}\left(alfa1 * g1\left(\beta 12^{\frac{5}{2}} - \gamma 12bar^{\frac{5}{2}} * \beta 22bar^{\frac{5}{2}}\right) + alfa2 * g2\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} +$$

$$\frac{4}{7}\,\text{alfa2}*\text{g3}\sqrt{\text{x3}}\,(\text{u3})^{\frac{5}{2}}+\frac{4}{7}\,\text{alfa1}*\text{g1}\sqrt{\text{x1}}\,(\text{u1})^{\frac{5}{2}}+$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^{2}\,\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)-\right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}}-\text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}}-\text{alfa2}*\text{g2}\right)\text{s2}+$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^{2}*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3}+\beta 22\text{bar}*\text{n2}*\text{Q2}-\beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right)-\right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)\left(\text{n3}+\beta 22\text{bar}\,\frac{\text{n2}*\text{Q2}}{\text{Q3}}-\beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3}-$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1}+\gamma 12\text{bar}\,\frac{\text{n3}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2}-\text{n3}\right)-\right.$$

$$\left.\beta 12^{1.5}*\text{n2}-\gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\,\frac{\text{n2}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}\,\beta 22\text{bar}\,\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$\text{V1}=-\text{d}\left(\frac{(\text{g1}\,\text{n1}-1)}{\text{n1}}\,\text{g1}\,\text{K11}+\frac{(\text{g3}\,\text{n3}-1)}{\text{n3}}\,\text{g3}\,\text{K33}+\right.$$

$$\left.\frac{(\text{g2}\,\text{n2}-1)}{\text{n2}}\,\text{g2}\,\text{K22}+2\,\text{g1}\,\text{g3}\,\text{K31}+\frac{4}{3}\,\text{g2}\,\text{g1}\,\text{K21}+\frac{4}{3}\,\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
Print["normx3 (computationally)=  ", normmx3];
Print["normx3 (by hand)=  ", normx3];
Print["normx1 (computationally)=  ", normmx1];
Print["normx1 (by hand)=  ", normx1];
Print["0 to x2 (h4 computationally)=  ", h4];
Print["0 to x2 (h4 by hand)=  ", AB];
```

```
    Print["x2 to x3 (h5 computationally)=  ", h5];
    Print["x2 to x3 (h5 by hand)=  ", CD];
    Print["x2 to x3 (h5 by handddddd)=  ", CCD];
    Print["x3 to x1 (h6 computationally)=  ", h6];
    Print["x3 to x1 (h6 by hand)=  ", EF];
    Print["K22  (computationally) = ", K222];
    Print["K22  (by hand) = ", K22];
    Print["K32  (computationally) = ", K332];
    Print["K32  (by hand) = ", K32];
    Print["K21  (computationally) = ", K221];
    Print["K21  (by hand) =", K21];
    Print["K31  (computationally) = ", K331];
    Print["K31  (by hand) = ", K31];
    Print["K33  (computationally) = ", K333];
    Print["K33  (by hand) = ", K33];
    Print["K11  (computationally) = ", K111];
    Print["K11  (by hand) = ", K11];
    Print["x2= ", x2];
    Print["x3= ", x3];
    Print["x1=  ", x1];
     Print["T by hand=", T1];
    Print["V by hand=", V1];
    Print["m2=", m2];
    Print["R=  ", R];
Print["Energywithoutmass = ", Energywithoutmass];
    Print["Energy=  ", Energy];
    If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
      Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
       H = 10^9 + 1;
       m2 = m2B;
       If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
   If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
   If[n == 3, Energy6bare = HX;
    Print["x1=", x1];
    Print["x3=", x3];
    Print["x2=", x2];
    Print["Kinetic Energy=", T1];
    Print["Potential Energy=", V1];
    Print["Volume Energy=", Volumeenergy];
   ]]];

(*Code for Plotting function *)
```

```
Plot1 = Plot[Evaluate[((y2[x] /.solution2) / x)^(3 / 2)],
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12 * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar * y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/ x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

```
R=  0.0462091341
normx2 (computationally)=  -0.607761033
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161279
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.24153562
0 to x2 (h4 by hand)=  7.2415349
x2 to x3 (h5 computationally)=  1.71191427
x2 to x3 (h5 by hand)=  1.71191689
x2 to x3 (h5 by handdddd)=  1.71191498
x3 to x1 (h6 computationally)=  4.70247063
x3 to x1 (h6 by hand)=  4.70247164
K22  (computationally) = 0.278217445
K22  (by hand) = 0.278217248
K32  (computationally) = 0.258478489
K32  (by hand) = 0.258478354
K21  (computationally) = 0.296870492
K21  (by hand) =0.296870473
K31  (computationally) = 0.294851536
K31  (by hand) = 0.294852227
K33  (computationally) = 0.244478671
K33  (by hand) = 0.244480527
K11  (computationally) = 0.510514742
K11  (by hand) = 0.510516889
x2= 0.120429145
x3= 0.137252069
x1=  0.25688802
T by hand=1275.78097
V by hand=-259.147402
m2=45
R=  0.0462091341
Energywithoutmass = 1955.82201
Energy=  42 195.822
R=  0.0462091341
```

```
normx2 (computationally)=  -0.607761033
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161276
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.24317376
0 to x2 (h4 by hand)=  7.24317304
x2 to x3 (h5 computationally)=  1.71230807
x2 to x3 (h5 by hand)=  1.71231065
x2 to x3 (h5 by handdddd)=  1.71230878
x3 to x1 (h6 computationally)=  4.70358311
x3 to x1 (h6 by hand)=  4.70358415
K22  (computationally) = 0.278248753
K22  (by hand) = 0.278248556
K32  (computationally) = 0.258507635
K32  (by hand) = 0.2585075
K21  (computationally) = 0.296904776
K21  (by hand) =0.296904758
K31  (computationally) = 0.294885583
K31  (by hand) = 0.294886261
K33  (computationally) = 0.244506237
K33  (by hand) = 0.24450806
K11  (computationally) = 0.510574297
K11  (by hand) = 0.510576507
x2= 0.120415522
x3= 0.137236458
x1=  0.256857807
T by hand=1276.07905
V by hand=-259.17726
m2=45.01
R=  0.0462091341
Energywithoutmass = 1955.75889
Energy=  42 195.7589
R=  0.0462091341
normx2 (computationally)=  -0.607761099
```

```
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161272
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.25791791
0 to x2 (h4 by hand)=  7.25791767
x2 to x3 (h5 computationally)=  1.71585165
x2 to x3 (h5 by hand)=  1.71585339
x2 to x3 (h5 by handdddd)=  1.71585236
x3 to x1 (h6 computationally)=  4.71359531
x3 to x1 (h6 by hand)=  4.71359639
K22  (computationally) = 0.27853061
K22  (by hand) = 0.278530545
K32  (computationally) = 0.258769941
K32  (by hand) = 0.258769897
K21  (computationally) = 0.297213158
K21  (by hand) =0.297213152
K31  (computationally) = 0.295191656
K31  (by hand) = 0.295192118
K33  (computationally) = 0.24475382
K33  (by hand) = 0.244755081
K11  (computationally) = 0.511109175
K11  (by hand) = 0.511111366
x2= 0.120293123
x3= 0.137096189
x1=  0.256586351
T by hand=1278.76163
V by hand=-259.445537
m2=45.1
R=  0.0462091341
Energywithoutmass = 1955.1997
Energy=  42 195.1997
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

```
R=  0.0462091341
normx2 (computationally)=  -0.607761106
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161254
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442451
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.27429982
0 to x2 (h4 by hand)=  7.27429963
x2 to x3 (h5 computationally)=  1.71979002
x2 to x3 (h5 by hand)=  1.71979141
x2 to x3 (h5 by handdddd)=  1.71979073
x3 to x1 (h6 computationally)=  4.72472084
x3 to x1 (h6 by hand)=  4.72472206
K22  (computationally) = 0.278843215
K22  (by hand) = 0.278843164
K32  (computationally) = 0.259060957
K32  (by hand) = 0.259060921
K21  (computationally) = 0.297555438
K21  (by hand) =0.297555433
K31  (computationally) = 0.295531595
K31  (by hand) = 0.295531964
K33  (computationally) = 0.245029158
K33  (by hand) = 0.245030166
K11  (computationally) = 0.511703193
K11  (by hand) = 0.511705618
x2= 0.120157554
x3= 0.136940838
x1=  0.256285728
T by hand=1281.74252
V by hand=-259.7435
m2=45.2
R=  0.0462091341
Energywithoutmass = 1954.59696
Energy=  42 194.597
R=  0.0462091341
```

```
normx2 (computationally)=  -0.60776112
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161254
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.29068214
0 to x2 (h4 by hand)=  7.29068206
x2 to x3 (h5 computationally)=  1.72372882
x2 to x3 (h5 by hand)=  1.72373004
x2 to x3 (h5 by handdddd)=  1.72372954
x3 to x1 (h6 computationally)=  4.73584692
x3 to x1 (h6 by hand)=  4.73584818
K22  (computationally) = 0.279155503
K22  (by hand) = 0.279155481
K32  (computationally) = 0.25935168
K32  (by hand) = 0.259351665
K21  (computationally) = 0.297897335
K21  (by hand) =0.297897333
K31  (computationally) = 0.295871225
K31  (by hand) = 0.295871551
K33  (computationally) = 0.245304416
K33  (by hand) = 0.245305312
K11  (computationally) = 0.512296335
K11  (by hand) = 0.51229882
x2= 0.120022441
x3= 0.136786014
x1=  0.255986141
T by hand=1284.72354
V by hand=-260.041128
m2=45.3
R=  0.0462091341
Energywithoutmass = 1954.01368
Energy=  42 194.0137
R=  0.0462091341
normx2 (computationally)=  -0.607761109
```

```
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161276
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.30706448
0 to x2 (h4 by hand)=  7.30706432
x2 to x3 (h5 computationally)=  1.72766878
x2 to x3 (h5 by hand)=  1.72767047
x2 to x3 (h5 by handdddd)=  1.7276695
x3 to x1 (h6 computationally)=  4.74697336
x3 to x1 (h6 by hand)=  4.74697442
K22  (computationally) = 0.279467356
K22  (by hand) = 0.279467312
K32  (computationally) = 0.25964206
K32  (by hand) = 0.25964203
K21  (computationally) = 0.298238839
K21  (by hand) =0.298238834
K31  (computationally) = 0.29621071
K31  (by hand) = 0.29621116
K33  (computationally) = 0.245580056
K33  (by hand) = 0.24558129
K11  (computationally) = 0.512888265
K11  (by hand) = 0.512890416
x2= 0.119887777
x3= 0.136631717
x1=  0.255687581
T by hand=1287.70469
V by hand=-260.338419
m2=45.4
R=  0.0462091341
Energywithoutmass = 1953.44965
Energy=  42 193.4496
R=  0.0462091341
normx2 (computationally)=  -0.607761138
normx2 (by hand)=  -0.607761206
```

```
normx3 (computationally)=  -0.543161261
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.3234476
0 to x2 (h4 by hand)=  7.32344765
x2 to x3 (h5 computationally)=  1.73160724
x2 to x3 (h5 by hand)=  1.73160837
x2 to x3 (h5 by handdddd)=  1.73160797
x3 to x1 (h6 computationally)=  4.75810104
x3 to x1 (h6 by hand)=  4.75810224
K22  (computationally) = 0.279779009
K22  (by hand) = 0.279779023
K32  (computationally) = 0.259932155
K32  (by hand) = 0.259932165
K21  (computationally) = 0.298579986
K21  (by hand) =0.298579988
K31  (computationally) = 0.296549442
K31  (by hand) = 0.296549744
K33  (computationally) = 0.245854278
K33  (by hand) = 0.245855119
K11  (computationally) = 0.513480295
K11  (by hand) = 0.51348266
x2= 0.119753567
x3= 0.136477926
x1=  0.255390051
T by hand=1290.68603
V by hand=-260.635352
m2=45.5
R=  0.0462091341
Energywithoutmass = 1952.90491
Energy=  42 192.9049
R=  0.0462091341
normx2 (computationally)=  -0.607761139
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161272
```

```
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442438
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.3398307
0 to x2 (h4 by hand)=  7.33983076
x2 to x3 (h5 computationally)=  1.735547
x2 to x3 (h5 by hand)=  1.73554828
x2 to x3 (h5 by handdddd)=  1.73554772
x3 to x1 (h6 computationally)=  4.76922863
x3 to x1 (h6 by hand)=  4.76922945
K22  (computationally) = 0.280090218
K22  (by hand) = 0.280090235
K32  (computationally) = 0.260221906
K32  (by hand) = 0.260221918
K21  (computationally) = 0.298920725
K21  (by hand) =0.298920726
K31  (computationally) = 0.296888046
K31  (by hand) = 0.296888391
K33  (computationally) = 0.246128979
K33  (by hand) = 0.246129938
K11  (computationally) = 0.514070344
K11  (by hand) = 0.514072021
x2= 0.119619801
x3= 0.136324657
x1=  0.25509353
T by hand=1293.66736
V by hand=-260.931753
m2=45.6
R=  0.0462091341
Energywithoutmass = 1952.37913
Energy=  42 192.3791
R=  0.0462091341
normx2 (computationally)=  -0.607761072
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161256
normx3 (by hand)=  -0.543161289
```

```
normx1 (computationally)=  -0.434442438
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.35621331
0 to x2 (h4 by hand)=  7.35621287
x2 to x3 (h5 computationally)=  1.73948754
x2 to x3 (h5 by hand)=  1.7394894
x2 to x3 (h5 by handdddd)=  1.73948827
x3 to x1 (h6 computationally)=  4.78035774
x3 to x1 (h6 by hand)=  4.78035872
K22  (computationally) = 0.28040084
K22  (by hand) = 0.28040072
K32  (computationally) = 0.260511197
K32  (by hand) = 0.260511114
K21  (computationally) = 0.299261107
K21  (by hand) =0.299261096
K31  (computationally) = 0.297226455
K31  (by hand) = 0.297226945
K33  (computationally) = 0.246403716
K33  (by hand) = 0.24640504
K11  (computationally) = 0.514661231
K11  (by hand) = 0.514663244
x2= 0.119486476
x3= 0.136171902
x1=  0.254798036
T by hand=1296.6491
V by hand=-261.228298
m2=45.7
R=  0.0462091341
Energywithoutmass = 1951.87215
Energy=  42 191.8722
R=  0.0462091341
normx2 (computationally)=  -0.607761137
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161272
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442439
```

```
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.37259779
0 to x2 (h4 by hand)=  7.37259784
x2 to x3 (h5 computationally)=  1.74342647
x2 to x3 (h5 by hand)=  1.74342778
x2 to x3 (h5 by handdddd)=  1.7434272
x3 to x1 (h6 computationally)=  4.79148714
x3 to x1 (h6 by hand)=  4.79148798
K22  (computationally) = 0.280711602
K22  (by hand) = 0.280711615
K32  (computationally) = 0.26080043
K32  (by hand) = 0.260800439
K21  (computationally) = 0.299601124
K21  (by hand) =0.299601125
K31  (computationally) = 0.297564072
K31  (by hand) = 0.297564423
K33  (computationally) = 0.246677133
K33  (by hand) = 0.24667811
K11  (computationally) = 0.515250604
K11  (by hand) = 0.515252314
x2= 0.119353602
x3= 0.136019647
x1=  0.254503541
T by hand=1299.6308
V by hand=-261.524091
m2=45.8
R=  0.0462091341
Energywithoutmass = 1951.38403
Energy=  42 191.384
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.0462091341
normx2 (computationally)=  -0.60776109
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161254
```

```
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.38898125
0 to x2 (h4 by hand)=  7.38898095
x2 to x3 (h5 computationally)=  1.74736687
x2 to x3 (h5 by hand)=  1.74736847
x2 to x3 (h5 by handdddd)=  1.7473676
x3 to x1 (h6 computationally)=  4.80261769
x3 to x1 (h6 by hand)=  4.80261869
K22  (computationally) = 0.281021617
K22  (by hand) = 0.281021533
K32  (computationally) = 0.261089121
K32  (by hand) = 0.261089064
K21  (computationally) = 0.299940763
K21  (by hand) =0.299940755
K31  (computationally) = 0.297901636
K31  (by hand) = 0.297902062
K33  (computationally) = 0.246950993
K33  (by hand) = 0.246952148
K11  (computationally) = 0.515840183
K11  (by hand) = 0.515842222
x2= 0.11922116
x3= 0.135867904
x1=  0.254210056
T by hand=1302.61283
V by hand=-261.819927
m2=45.9
R=  0.0462091341
Energywithoutmass = 1950.91439
Energy=  42190.9144
R=  0.0462091341
normx2 (computationally)=  -0.607761115
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161277
normx3 (by hand)=  -0.543161289
```

```
normx1 (computationally)=  -0.434442441
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.40536587
0 to x2 (h4 by hand)=  7.40536576
x2 to x3 (h5 computationally)=  1.75130702
x2 to x3 (h5 by hand)=  1.75130868
x2 to x3 (h5 by handdddd)=  1.75130776
x3 to x1 (h6 computationally)=  4.81374845
x3 to x1 (h6 by hand)=  4.81374929
K22  (computationally) = 0.281331556
K22  (by hand) = 0.281331524
K32  (computationally) = 0.26137766
K32  (by hand) = 0.261377638
K21  (computationally) = 0.30028003
K21  (by hand) =0.300280027
K31  (computationally) = 0.298238738
K31  (by hand) = 0.298239181
K33  (computationally) = 0.247224422
K33  (by hand) = 0.24722564
K11  (computationally) = 0.516428286
K11  (by hand) = 0.51643003
x2= 0.11908916
x3= 0.135716664
x1=  0.253917561
T by hand=1305.59489
V by hand=-262.115205
m2=46.
R=  0.0462091341
Energywithoutmass = 1950.46321
Energy=  42 190.4632
R=  0.0462091341
normx2 (computationally)=  -0.607761086
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161254
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.43444244
```

```
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.42175016
0 to x2 (h4 by hand)=  7.42174982
x2 to x3 (h5 computationally)=  1.75524725
x2 to x3 (h5 by hand)=  1.75524891
x2 to x3 (h5 by handdddd)=  1.75524798
x3 to x1 (h6 computationally)=  4.82488048
x3 to x1 (h6 by hand)=  4.82488153
K22  (computationally) = 0.28164096
K22  (by hand) = 0.281640867
K32  (computationally) = 0.261665749
K32  (by hand) = 0.261665686
K21  (computationally) = 0.300618934
K21  (by hand) =0.300618925
K31  (computationally) = 0.29857546
K31  (by hand) = 0.298575902
K33  (computationally) = 0.24749739
K33  (by hand) = 0.247498587
K11  (computationally) = 0.517016665
K11  (by hand) = 0.5170188
x2= 0.118957592
x3= 0.135565921
x1=  0.253626064
T by hand=1308.57721
V by hand=-262.410359
m2=46.1
R=  0.0462091341
Energywithoutmass = 1950.03032
Energy=  42190.0303
R=  0.0462091341
normx2 (computationally)=  -0.607761141
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161266
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442451
normx1 (by hand)=  -0.434442453
```

```
0 to x2 (h4 computationally)=  7.43813575
0 to x2 (h4 by hand)=  7.43813583
x2 to x3 (h5 computationally)=  1.75918685
x2 to x3 (h5 by hand)=  1.75918804
x2 to x3 (h5 by handdddd)=  1.75918759
x3 to x1 (h6 computationally)=  4.83601303
x3 to x1 (h6 by hand)=  4.83601417
K22  (computationally) = 0.281950336
K22  (by hand) = 0.281950358
K32  (computationally) = 0.261953706
K32  (by hand) = 0.261953721
K21  (computationally) = 0.30095748
K21  (by hand) =0.300957482
K31  (computationally) = 0.298911644
K31  (by hand) = 0.298911964
K33  (computationally) = 0.247769676
K33  (by hand) = 0.24777057
K11  (computationally) = 0.517604094
K11  (by hand) = 0.517606344
x2= 0.118826462
x3= 0.135415674
x1=  0.25333555
T by hand=1311.55963
V by hand=-262.70505
m2=46.2
R=  0.0462091341
Energywithoutmass = 1949.61563
Energy=  42189.6156
R=  0.0462091341
normx2 (computationally)=  -0.607761087
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161266
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.45452036
```

```
0 to x2 (h4 by hand)=  7.45452003
x2 to x3 (h5 computationally)=  1.76312838
x2 to x3 (h5 by hand)=  1.76313024
x2 to x3 (h5 by handdddd)=  1.76312913
x3 to x1 (h6 computationally)=  4.84714623
x3 to x1 (h6 by hand)=  4.84714741
K22  (computationally) = 0.282258982
K22  (by hand) = 0.282258893
K32  (computationally) = 0.262241149
K32  (by hand) = 0.262241088
K21  (computationally) = 0.301295638
K21  (by hand) =0.30129563
K31  (computationally) = 0.299247876
K31  (by hand) = 0.299248367
K33  (computationally) = 0.248042756
K33  (by hand) = 0.248044091
K11  (computationally) = 0.518190897
K11  (by hand) = 0.518193307
x2= 0.118695755
x3= 0.135265931
x1=  0.253046017
T by hand=1314.54227
V by hand=-262.999632
m2=46.3
R=  0.0462091341
Energywithoutmass = 1949.21882
Energy=  42189.2188
R=  0.0462091341
normx2 (computationally)=  -0.607761134
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161266
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.47090647
0 to x2 (h4 by hand)=  7.4709065
```

```
x2 to x3 (h5 computationally)=  1.76706812
x2 to x3 (h5 by hand)=  1.7670694
x2 to x3 (h5 by handdddd)=  1.76706886
x3 to x1 (h6 computationally)=  4.85827957
x3 to x1 (h6 by hand)=  4.85828046
K22  (computationally) = 0.282567661
K22  (by hand) = 0.282567669
K32  (computationally) = 0.262528459
K32  (by hand) = 0.262528465
K21  (computationally) = 0.301633423
K21  (by hand) =0.301633423
K31  (computationally) = 0.299583265
K31  (by hand) = 0.299583609
K33  (computationally) = 0.248314328
K33  (by hand) = 0.248315282
K11  (computationally) = 0.518775898
K11  (by hand) = 0.518777696
x2= 0.118565483
x3= 0.135116669
x1=  0.252757447
T by hand=1317.52478
V by hand=-263.293327
m2=46.4
R=  0.0462091341
Energywithoutmass = 1948.84005
Energy=  42188.84
R=  0.0462091341
normx2 (computationally)=  -0.607761141
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161274
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.43444244
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.48729239
0 to x2 (h4 by hand)=  7.48729247
x2 to x3 (h5 computationally)=  1.77100896
```

```
x2 to x3 (h5 by hand)=  1.77101028
x2 to x3 (h5 by handdddd)=  1.77100971
x3 to x1 (h6 computationally)=  4.86941404
x3 to x1 (h6 by hand)=  4.8694149
K22  (computationally) = 0.28287586
K22  (by hand) = 0.282875882
K32  (computationally) = 0.262815395
K32  (by hand) = 0.26281541
K21  (computationally) = 0.301970855
K21  (by hand) =0.301970857
K31  (computationally) = 0.29991854
K31  (by hand) = 0.299918896
K33  (computationally) = 0.24858623
K33  (by hand) = 0.248587222
K11  (computationally) = 0.51936116
K11  (by hand) = 0.519362921
x2= 0.118435634
x3= 0.134967903
x1=  0.252469854
T by hand=1320.50764
V by hand=-263.587085
m2=46.5
R=  0.0462091341
Energywithoutmass = 1948.47891
Energy=  42188.4789
R=  0.0462091341
normx2 (computationally)=  -0.607761108
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161269
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442443
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.50367815
0 to x2 (h4 by hand)=  7.50367798
x2 to x3 (h5 computationally)=  1.77495039
x2 to x3 (h5 by hand)=  1.77495204
```

```
x2 to x3 (h5 by handdddd)=  1.77495114
x3 to x1 (h6 computationally)=  4.88054939
x3 to x1 (h6 by hand)=  4.88055037
K22  (computationally) = 0.283183583
K22  (by hand) = 0.283183537
K32  (computationally) = 0.26310194
K32  (by hand) = 0.263101908
K21  (computationally) = 0.302307923
K21  (by hand) =0.302307919
K31  (computationally) = 0.300253563
K31  (by hand) = 0.300254003
K33  (computationally) = 0.24885809
K33  (by hand) = 0.248859297
K11  (computationally) = 0.519946163
K11  (by hand) = 0.519948156
x2= 0.118306205
x3= 0.134819625
x1=  0.252183229
T by hand=1323.49072
V by hand=-263.880672
m2=46.6
R=  0.0462091341
Energywithoutmass = 1948.13532
Energy=  42188.1353
R=  0.0462091341
normx2 (computationally)=  -0.607761035
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161273
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.52006372
0 to x2 (h4 by hand)=  7.520063
x2 to x3 (h5 computationally)=  1.778893
x2 to x3 (h5 by hand)=  1.77889564
x2 to x3 (h5 by handdddd)=  1.77889375
```

```
x3 to x1 (h6 computationally)=  4.89168504
x3 to x1 (h6 by hand)=  4.89168587
K22  (computationally) = 0.283490827
K22  (by hand) = 0.283490628
K32  (computationally) = 0.263388115
K32  (by hand) = 0.263387979
K21  (computationally) = 0.302644608
K21  (by hand) =0.302644589
K31  (computationally) = 0.300588457
K31  (by hand) = 0.300589154
K33  (computationally) = 0.249130334
K33  (by hand) = 0.249132206
K11  (computationally) = 0.520529906
K11  (by hand) = 0.520531712
x2= 0.118177195
x3= 0.134671837
x1=  0.251897557
T by hand=1326.47389
V by hand=-264.173907
m2=46.7
R=  0.0462091341
Energywithoutmass = 1947.80913
Energy=  42187.8091
R=  0.0462091341
normx2 (computationally)=  -0.607761047
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161273
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.53645062
0 to x2 (h4 by hand)=  7.53644999
x2 to x3 (h5 computationally)=  1.78283396
x2 to x3 (h5 by hand)=  1.78283646
x2 to x3 (h5 by handdddd)=  1.78283472
x3 to x1 (h6 computationally)=  4.90282162
```

```
x3 to x1 (h6 by hand)=  4.90282245
K22  (computationally) = 0.283798055
K22  (by hand) = 0.283797882
K32  (computationally) = 0.263674131
K32  (by hand) = 0.263674012
K21  (computationally) = 0.302980953
K21  (by hand) =0.302980937
K31  (computationally) = 0.300922579
K31  (by hand) = 0.300923238
K33  (computationally) = 0.249401156
K33  (by hand) = 0.249402931
K11  (computationally) = 0.521113305
K11  (by hand) = 0.521115104
x2= 0.118048609
x3= 0.134524521
x1=  0.251612839
T by hand=1329.45718
V by hand=-264.46668
m2=46.8
R=  0.0462091341
Energywithoutmass = 1947.50039
Energy=  42187.5004
R=  0.0462091341
normx2 (computationally)=  -0.607761047
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.54316127
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.43444244
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.55283767
0 to x2 (h4 by hand)=  7.55283704
x2 to x3 (h5 computationally)=  1.78677521
x2 to x3 (h5 by hand)=  1.78677766
x2 to x3 (h5 by handdddd)=  1.78677597
x3 to x1 (h6 computationally)=  4.91395901
x3 to x1 (h6 by hand)=  4.91395994
```

```
K22  (computationally) = 0.284104912
K22  (by hand) = 0.284104738
K32  (computationally) = 0.263959817
K32  (by hand) = 0.263959698
K21  (computationally) = 0.303316941
K21  (by hand) =0.303316925
K31  (computationally) = 0.301256385
K31  (by hand) = 0.301257031
K33  (computationally) = 0.24967178
K33  (by hand) = 0.249673521
K11  (computationally) = 0.521696338
K11  (by hand) = 0.521698306
x2= 0.117920439
x3= 0.134377685
x1=  0.251329072
T by hand=1332.44067
V by hand=-264.759228
m2=46.9
R=  0.0462091341
Energywithoutmass = 1947.20883
Energy=  42187.2088
R=  0.0462091341
normx2 (computationally)=  -0.607761047
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161277
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.56922501
0 to x2 (h4 by hand)=  7.56922438
x2 to x3 (h5 computationally)=  1.79071687
x2 to x3 (h5 by hand)=  1.79071944
x2 to x3 (h5 by handdddd)=  1.79071763
x3 to x1 (h6 computationally)=  4.92509723
x3 to x1 (h6 by hand)=  4.92509833
K22  (computationally) = 0.28441144
```

```
K22  (by hand) = 0.284411266
K32  (computationally) = 0.264245207
K32  (by hand) = 0.264245088
K21  (computationally) = 0.303652577
K21  (by hand) =0.303652561
K31  (computationally) = 0.301589905
K31  (by hand) = 0.301590584
K33  (computationally) = 0.249942307
K33  (by hand) = 0.249944138
K11  (computationally) = 0.52227904
K11  (by hand) = 0.522281349
x2= 0.117792684
x3= 0.134231327
x1=  0.251046249
T by hand=1335.42437
V by hand=-265.051591
m2=47.
R=  0.0462091341
Energywithoutmass = 1946.93429
Energy=  42186.9343
R=  0.0462091341
normx2 (computationally)=  -0.607761029
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161263
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442438
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.58561243
0 to x2 (h4 by hand)=  7.58561166
x2 to x3 (h5 computationally)=  1.79465849
x2 to x3 (h5 by hand)=  1.79466108
x2 to x3 (h5 by handdddd)=  1.79465925
x3 to x1 (h6 computationally)=  4.93623563
x3 to x1 (h6 by hand)=  4.93623659
K22  (computationally) = 0.284717574
K22  (by hand) = 0.284717361
```

```
K32  (computationally) = 0.264530242
K32  (by hand) = 0.264530096
K21  (computationally) = 0.303987827
K21  (by hand) =0.303987807
K31  (computationally) = 0.301923
K31  (by hand) = 0.301923681
K33  (computationally) = 0.250212384
K33  (by hand) = 0.250214209
K11  (computationally) = 0.522860141
K11  (by hand) = 0.522862194
x2= 0.117665338
x3= 0.13408544
x1=  0.250764356
T by hand=1338.40804
V by hand=-265.34332
m2=47.1
R=  0.0462091341
Energywithoutmass = 1946.67674
Energy=  42186.6767
R=  0.0462091341
normx2 (computationally)=  -0.60776114
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161264
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.43444244
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.60200173
0 to x2 (h4 by hand)=  7.60200181
x2 to x3 (h5 computationally)=  1.79859806
x2 to x3 (h5 by hand)=  1.79859927
x2 to x3 (h5 by handdddd)=  1.79859883
x3 to x1 (h6 computationally)=  4.94737496
x3 to x1 (h6 by hand)=  4.94737595
K22  (computationally) = 0.285023867
K22  (by hand) = 0.285023889
K32  (computationally) = 0.264815222
```

```
K32  (by hand) = 0.264815237
K21  (computationally) = 0.304322755
K21  (by hand) =0.304322757
K31  (computationally) = 0.302255242
K31  (by hand) = 0.302255567
K33  (computationally) = 0.250480846
K33  (by hand) = 0.250481754
K11  (computationally) = 0.523440954
K11  (by hand) = 0.523442921
x2= 0.117538412
x3= 0.133940014
x1=  0.250483395
T by hand=1341.39182
V by hand=-265.634578
m2=47.2
R=  0.0462091341
Energywithoutmass = 1946.43613
Energy=  42186.4361
R=  0.0462091341
normx2 (computationally)=  -0.607761087
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.54316126
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442449
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.61838932
0 to x2 (h4 by hand)=  7.61838899
x2 to x3 (h5 computationally)=  1.80254087
x2 to x3 (h5 by hand)=  1.80254268
x2 to x3 (h5 by handdddd)=  1.80254164
x3 to x1 (h6 computationally)=  4.9585153
x3 to x1 (h6 by hand)=  4.95851651
K22  (computationally) = 0.285329222
K22  (by hand) = 0.285329131
K32  (computationally) = 0.265099597
K32  (by hand) = 0.265099535
```

K21 (computationally) = 0.304657313

K21 (by hand) =0.304657304

K31 (computationally) = 0.302587873

K31 (by hand) = 0.302588352

K33 (computationally) = 0.250750949

K33 (by hand) = 0.25075225

K11 (computationally) = 0.524021982

K11 (by hand) = 0.52402444

x2= 0.117411882

x3= 0.133795073

x1= 0.250203372

T by hand=1344.37602

V by hand=−265.92614

m2=47.3

R= 0.0462091341

Energywithoutmass = 1946.21194

Energy= 42 186.2119

R= 0.0462091341

normx2 (computationally)= −0.607761083

normx2 (by hand)= −0.607761206

normx3 (computationally)= −0.543161253

normx3 (by hand)= −0.543161289

normx1 (computationally)= −0.434442437

normx1 (by hand)= −0.434442453

0 to x2 (h4 computationally)= 7.63477779

0 to x2 (h4 by hand)= 7.63477743

x2 to x3 (h5 computationally)= 1.80648281

x2 to x3 (h5 by hand)= 1.80648457

x2 to x3 (h5 by handdddd)= 1.80648358

x3 to x1 (h6 computationally)= 4.96965569

x3 to x1 (h6 by hand)= 4.96965674

K22 (computationally) = 0.285634436

K22 (by hand) = 0.285634337

K32 (computationally) = 0.265383759

K32 (by hand) = 0.26538369

K21 (computationally) = 0.304991498

```
K21  (by hand) =0.304991489
K31  (computationally) = 0.302919887
K31  (by hand) = 0.302920354
K33  (computationally) = 0.25102013
K33  (by hand) = 0.251021395
K11  (computationally) = 0.524601176
K11  (by hand) = 0.524603329
x2= 0.117285761
x3= 0.133650593
x1=  0.249924259
T by hand=1347.36012
V by hand=-266.216923
m2=47.4
R=  0.0462091341
Energywithoutmass = 1946.00442
Energy=  42186.0044
R=  0.0462091341
normx2 (computationally)=  -0.607761126
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161277
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.65116715
0 to x2 (h4 by hand)=  7.65116712
x2 to x3 (h5 computationally)=  1.81042471
x2 to x3 (h5 by hand)=  1.81042631
x2 to x3 (h5 by handdddd)=  1.81042549
x3 to x1 (h6 computationally)=  4.98079663
x3 to x1 (h6 by hand)=  4.98079743
K22  (computationally) = 0.285939508
K22  (by hand) = 0.285939501
K32  (computationally) = 0.265667736
K32  (by hand) = 0.265667731
K21  (computationally) = 0.305325338
K21  (by hand) =0.305325338
```

K31 (computationally) = 0.303251517

K31 (by hand) = 0.303251946

K33 (computationally) = 0.251288999

K33 (by hand) = 0.251290187

K11 (computationally) = 0.525179581

K11 (by hand) = 0.525181253

x2= 0.117160046

x3= 0.133506578

x1= 0.249646063

T by hand=1350.34434

V by hand=−266.507363

m2=47.5

R= 0.0462091341

Energywithoutmass = 1945.81333

Energy= 42 185.8133

R= 0.0462091341

normx2 (computationally)= −0.607761028

normx2 (by hand)= −0.607761206

normx3 (computationally)= −0.543161253

normx3 (by hand)= −0.543161289

normx1 (computationally)= −0.434442439

normx1 (by hand)= −0.434442453

0 to x2 (h4 computationally)= 7.66755505

0 to x2 (h4 by hand)= 7.66755427

x2 to x3 (h5 computationally)= 1.81436834

x2 to x3 (h5 by hand)= 1.81437081

x2 to x3 (h5 by handdddd)= 1.81436912

x3 to x1 (h6 computationally)= 4.99193906

x3 to x1 (h6 by hand)= 4.99194015

K22 (computationally) = 0.286243726

K22 (by hand) = 0.28624351

K32 (computationally) = 0.265951111

K32 (by hand) = 0.265950962

K21 (computationally) = 0.305658827

K21 (by hand) =0.305658807

K31 (computationally) = 0.303583187

```
K31  (by hand) = 0.303583837
K33  (computationally) = 0.251558459
K33  (by hand) = 0.251560195
K11  (computationally) = 0.525758908
K11  (by hand) = 0.525761203
x2= 0.117034723
x3= 0.133363029
x1=  0.249368791
T by hand=1353.329
V by hand=-266.79807
m2=47.6
R=  0.0462091341
Energywithoutmass = 1945.63838
Energy=  42 185.6384
R=  0.0462091341
normx2 (computationally)=  -0.607761113
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161268
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.68394551
0 to x2 (h4 by hand)=  7.68394538
x2 to x3 (h5 computationally)=  1.81830945
x2 to x3 (h5 by hand)=  1.81831108
x2 to x3 (h5 by handdddd)=  1.81831023
x3 to x1 (h6 computationally)=  5.00308143
x3 to x1 (h6 by hand)=  5.00308234
K22  (computationally) = 0.286548314
K22  (by hand) = 0.286548278
K32  (computationally) = 0.266234559
K32  (by hand) = 0.266234535
K21  (computationally) = 0.305991973
K21  (by hand) =0.305991969
K31  (computationally) = 0.303913867
K31  (by hand) = 0.303914304
```

K33 (computationally) = 0.251826074

K33 (by hand) = 0.251827273

K11 (computationally) = 0.526336153

K11 (by hand) = 0.526338018

x2= 0.116909812

x3= 0.133219928

x1= 0.249092414

T by hand=1356.31348

V by hand=−267.087772

m2=47.7

R= 0.0462091341

Energywithoutmass = 1945.47985

Energy= 42 185.4799

R= 0.0462091341

normx2 (computationally)= −0.607761028

normx2 (by hand)= −0.607761206

normx3 (computationally)= −0.543161263

normx3 (by hand)= −0.543161289

normx1 (computationally)= −0.434442437

normx1 (by hand)= −0.434442453

0 to x2 (h4 computationally)= 7.70033415

0 to x2 (h4 by hand)= 7.70033335

x2 to x3 (h5 computationally)= 1.82225358

x2 to x3 (h5 by hand)= 1.82225623

x2 to x3 (h5 by handdddd)= 1.82225437

x3 to x1 (h6 computationally)= 5.01422484

x3 to x1 (h6 by hand)= 5.0142258

K22 (computationally) = 0.28685194

K22 (by hand) = 0.286851722

K32 (computationally) = 0.266517379

K32 (by hand) = 0.26651723

K21 (computationally) = 0.306324751

K21 (by hand) =0.306324731

K31 (computationally) = 0.304244888

K31 (by hand) = 0.304245585

K33 (computationally) = 0.252095164

```
K33  (by hand) = 0.25209703
K11  (computationally) = 0.52691367
K11  (by hand) = 0.52691574
x2= 0.116785286
x3= 0.133077297
x1=  0.248816947
T by hand=1359.29835
V by hand=-267.377753
m2=47.8
R=  0.0462091341
Energywithoutmass = 1945.33714
Energy=  42 185.3371
R=  0.0462091341
normx2 (computationally)=  -0.607761027
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161274
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442442
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.71672413
0 to x2 (h4 by hand)=  7.71672333
x2 to x3 (h5 computationally)=  1.82619656
x2 to x3 (h5 by hand)=  1.8261994
x2 to x3 (h5 by handdddd)=  1.82619735
x3 to x1 (h6 computationally)=  5.02536879
x3 to x1 (h6 by hand)=  5.02536973
K22  (computationally) = 0.28715557
K22  (by hand) = 0.287155351
K32  (computationally) = 0.266800075
K32  (by hand) = 0.266799925
K21  (computationally) = 0.306657193
K21  (by hand) =0.306657172
K31  (computationally) = 0.304575261
K31  (by hand) = 0.304576008
K33  (computationally) = 0.252363212
K33  (by hand) = 0.252365218
```

```
K11  (computationally) = 0.527490291
K11  (by hand) = 0.527492333
x2= 0.11666116
x3= 0.132935114
x1=  0.248542375
T by hand=1362.28327
V by hand=-267.667208
m2=47.9
R=  0.0462091341
Energywithoutmass = 1945.21046
Energy=  42185.2105
R=  0.0462091341
normx2 (computationally)=  -0.607761027
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161255
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442453
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.7331144
0 to x2 (h4 by hand)=  7.7331136
x2 to x3 (h5 computationally)=  1.83013891
x2 to x3 (h5 by hand)=  1.83014145
x2 to x3 (h5 by handdddd)=  1.8301397
x3 to x1 (h6 computationally)=  5.03651409
x3 to x1 (h6 by hand)=  5.03651545
K22  (computationally) = 0.287458883
K22  (by hand) = 0.287458664
K32  (computationally) = 0.267082448
K32  (by hand) = 0.267082297
K21  (computationally) = 0.306989309
K21  (by hand) =0.306989288
K31  (computationally) = 0.304905123
K31  (by hand) = 0.304905791
K33  (computationally) = 0.252630424
K33  (by hand) = 0.252632212
K11  (computationally) = 0.528067436
```

```
K11  (by hand) = 0.528070239
x2= 0.116537427
x3= 0.132793376
x1=  0.248268704
T by hand=1365.26848
V by hand=-267.956548
m2=48.
R=  0.0462091341
Energywithoutmass = 1945.09964
Energy=  42185.0996
R=  0.0462091341
normx2 (computationally)=  -0.607761108
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161274
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.74950596
0 to x2 (h4 by hand)=  7.7495058
x2 to x3 (h5 computationally)=  1.83408072
x2 to x3 (h5 by hand)=  1.83408254
x2 to x3 (h5 by handdddd)=  1.83408152
x3 to x1 (h6 computationally)=  5.0476586
x3 to x1 (h6 by hand)=  5.04765946
K22  (computationally) = 0.287762188
K22  (by hand) = 0.287762142
K32  (computationally) = 0.267364711
K32  (by hand) = 0.26736468
K21  (computationally) = 0.307321039
K21  (by hand) =0.307321035
K31  (computationally) = 0.305234441
K31  (by hand) = 0.305234926
K33  (computationally) = 0.252897041
K33  (by hand) = 0.252898372
K11  (computationally) = 0.528641359
K11  (by hand) = 0.528643138
```

```
x2= 0.116414091
x3= 0.132652087
x1=  0.247995899
T by hand=1368.25339
V by hand=-268.24481
m2=48.1
R=  0.0462091341
Energywithoutmass = 1945.00463
Energy=  42185.0046
R=  0.0462091341
normx2 (computationally)=  -0.607761139
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161254
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.43444245
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.76589721
0 to x2 (h4 by hand)=  7.76589729
x2 to x3 (h5 computationally)=  1.83802264
x2 to x3 (h5 by hand)=  1.83802372
x2 to x3 (h5 by handdddd)=  1.83802344
x3 to x1 (h6 computationally)=  5.05880532
x3 to x1 (h6 by hand)=  5.05880666
K22  (computationally) = 0.288064991
K22  (by hand) = 0.288065014
K32  (computationally) = 0.267646554
K32  (by hand) = 0.267646569
K21  (computationally) = 0.307652471
K21  (by hand) =0.307652473
K31  (computationally) = 0.305563453
K31  (by hand) = 0.305563745
K33  (computationally) = 0.253163253
K33  (by hand) = 0.253164069
K11  (computationally) = 0.529217428
K11  (by hand) = 0.529220049
x2= 0.116291139
```

```
x3= 0.132511242
x1=  0.247723998
T by hand=1371.23887
V by hand=-268.533494
m2=48.2
R=  0.0462091341
Energywithoutmass = 1944.92514
Energy=  42 184.9251
R=  0.0462091341
normx2 (computationally)=  -0.607761055
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161269
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.78228728
0 to x2 (h4 by hand)=  7.78228669
x2 to x3 (h5 computationally)=  1.84196796
x2 to x3 (h5 by hand)=  1.84197039
x2 to x3 (h5 by handdddd)=  1.84196876
x3 to x1 (h6 computationally)=  5.06995129
x3 to x1 (h6 by hand)=  5.06995221
K22  (computationally) = 0.288367037
K22  (by hand) = 0.288366876
K32  (computationally) = 0.267927921
K32  (by hand) = 0.26792781
K21  (computationally) = 0.307983489
K21  (by hand) =0.307983474
K31  (computationally) = 0.305892854
K31  (by hand) = 0.305893495
K33  (computationally) = 0.253431335
K33  (by hand) = 0.253433065
K11  (computationally) = 0.529790664
K11  (by hand) = 0.529792628
x2= 0.116168567
x3= 0.132370861
```

```
x1=  0.247452962
T by hand=1374.22427
V by hand=-268.821697
m2=48.3
R=  0.0462091341
Energywithoutmass = 1944.86093
Energy=  42 184.8609
R=  0.0462091341
normx2 (computationally)=  -0.607761139
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161273
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442452
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.79867976
0 to x2 (h4 by hand)=  7.79867985
x2 to x3 (h5 computationally)=  1.84590971
x2 to x3 (h5 by hand)=  1.84591112
x2 to x3 (h5 by handdddd)=  1.84591051
x3 to x1 (h6 computationally)=  5.08109895
x3 to x1 (h6 by hand)=  5.08110012
K22  (computationally) = 0.288669418
K22  (by hand) = 0.288669441
K32  (computationally) = 0.268209303
K32  (by hand) = 0.268209319
K21  (computationally) = 0.308314231
K21  (by hand) =0.308314233
K31  (computationally) = 0.306221075
K31  (by hand) = 0.306221455
K33  (computationally) = 0.25369679
K33  (by hand) = 0.253697848
K11  (computationally) = 0.530364882
K11  (by hand) = 0.530367207
x2= 0.11604639
x3= 0.132230902
x1=  0.247182808
```

```
T by hand=1377.20993
V by hand=-269.109573
m2=48.4
R=  0.0462091341
Energywithoutmass = 1944.8123
Energy=  42 184.8123
R=  0.0462091341
normx2 (computationally)=  -0.607761139
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161268
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442439
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.81507147
0 to x2 (h4 by hand)=  7.81507155
x2 to x3 (h5 computationally)=  1.84985304
x2 to x3 (h5 by hand)=  1.84985437
x2 to x3 (h5 by handdddd)=  1.84985385
x3 to x1 (h6 computationally)=  5.09224657
x3 to x1 (h6 by hand)=  5.09224754
K22  (computationally) = 0.288971162
K22  (by hand) = 0.288971185
K32  (computationally) = 0.268490227
K32  (by hand) = 0.268490243
K21  (computationally) = 0.308644586
K21  (by hand) =0.308644588
K31  (computationally) = 0.306549269
K31  (by hand) = 0.306549626
K33  (computationally) = 0.253962858
K33  (by hand) = 0.253963854
K11  (computationally) = 0.530937332
K11  (by hand) = 0.530939287
x2= 0.115924591
x3= 0.13209139
x1=  0.246913515
T by hand=1380.19558
```

V by hand=-269.39699

m2=48.5

R= 0.0462091341

Energywithoutmass = 1944.77885

Energy= 42184.7788

R= 0.0462091341

normx2 (computationally)= -0.607761116

normx2 (by hand)= -0.607761206

normx3 (computationally)= -0.543161271

normx3 (by hand)= -0.543161289

normx1 (computationally)= -0.434442437

normx1 (by hand)= -0.434442453

0 to x2 (h4 computationally)= 7.83146316

0 to x2 (h4 by hand)= 7.83146306

x2 to x3 (h5 computationally)= 1.85379723

x2 to x3 (h5 by hand)= 1.85379892

x2 to x3 (h5 by handdddd)= 1.85379804

x3 to x1 (h6 computationally)= 5.10339503

x3 to x1 (h6 by hand)= 5.10339593

K22 (computationally) = 0.289272505

K22 (by hand) = 0.289272477

K32 (computationally) = 0.268770826

K32 (by hand) = 0.268770806

K21 (computationally) = 0.308974606

K21 (by hand) =0.308974603

K31 (computationally) = 0.306877304

K31 (by hand) = 0.306877756

K33 (computationally) = 0.254229123

K33 (by hand) = 0.25423037

K11 (computationally) = 0.531509616

K11 (by hand) = 0.531511467

x2= 0.11580317

x3= 0.13195232

x1= 0.24664509

T by hand=1383.18148

V by hand=-269.684329

```
m2=48.6
R=  0.0462091341
Energywithoutmass = 1944.76045
Energy=  42 184.7605
R=  0.0462091341
normx2 (computationally)=  -0.607761085
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161275
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442448
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.84785504
0 to x2 (h4 by hand)=  7.84785469
x2 to x3 (h5 computationally)=  1.8577417
x2 to x3 (h5 by hand)=  1.85774387
x2 to x3 (h5 by handdddd)=  1.85774251
x3 to x1 (h6 computationally)=  5.11454454
x3 to x1 (h6 by hand)=  5.11454563
K22  (computationally) = 0.28957351
K22  (by hand) = 0.289573414
K32  (computationally) = 0.269051122
K32  (by hand) = 0.269051056
K21  (computationally) = 0.3093043
K21  (by hand) =0.309304291
K31  (computationally) = 0.307205054
K31  (by hand) = 0.307205629
K33  (computationally) = 0.254495214
K33  (by hand) = 0.25449678
K11  (computationally) = 0.532082072
K11  (by hand) = 0.532084316
x2= 0.115682128
x3= 0.131813686
x1=  0.246377531
T by hand=1386.16765
V by hand=-269.971607
m2=48.7
```

```
R=  0.0462091341
Energywithoutmass = 1944.75704
Energy=  42184.757
R=  0.0462091341
normx2 (computationally)=  -0.607761087
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161274
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442437
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.86424761
0 to x2 (h4 by hand)=  7.86424728
x2 to x3 (h5 computationally)=  1.86168549
x2 to x3 (h5 by hand)=  1.86168761
x2 to x3 (h5 by handdddd)=  1.8616863
x3 to x1 (h6 computationally)=  5.1256941
x3 to x1 (h6 by hand)=  5.12569499
K22  (computationally) = 0.289874331
K22  (by hand) = 0.28987424
K32  (computationally) = 0.269331187
K32  (by hand) = 0.269331124
K21  (computationally) = 0.309633635
K21  (by hand) =0.309633626
K31  (computationally) = 0.307532248
K31  (by hand) = 0.307532811
K33  (computationally) = 0.254760509
K33  (by hand) = 0.254762042
K11  (computationally) = 0.53265276
K11  (by hand) = 0.532654634
x2= 0.115561465
x3= 0.13167548
x1=  0.246110817
T by hand=1389.15373
V by hand=-270.25815
m2=48.8
R=  0.0462091341
```

```
Energywithoutmass = 1944.76863
Energy=  42 184.7686
R=  0.0462091341
normx2 (computationally)=  -0.607761139
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161276
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442451
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.84949497
0 to x2 (h4 by hand)=  7.84949505
x2 to x3 (h5 computationally)=  1.85813499
x2 to x3 (h5 by hand)=  1.85813647
x2 to x3 (h5 by handdddd)=  1.85813581
x3 to x1 (h6 computationally)=  5.11565957
x3 to x1 (h6 by hand)=  5.11566071
K22  (computationally) = 0.289603814
K22  (by hand) = 0.289603838
K32  (computationally) = 0.269079244
K32  (by hand) = 0.26907926
K21  (computationally) = 0.309337262
K21  (by hand) =0.309337264
K31  (computationally) = 0.307237539
K31  (by hand) = 0.307237935
K33  (computationally) = 0.254521085
K33  (by hand) = 0.254522191
K11  (computationally) = 0.532139276
K11  (by hand) = 0.532141543
x2= 0.115670048
x3= 0.131799841
x1=  0.246350821
T by hand=1386.46623
V by hand=-270.000173
m2=48.71
R=  0.0462091341
Energywithoutmass = 1944.7576
```

```
Energy=  42 184.7576
R=  0.0462091341
normx2 (computationally)=  −0.607761131
normx2 (by hand)=  −0.607761206
normx3 (computationally)=  −0.543161253
normx3 (by hand)=  −0.543161289
normx1 (computationally)=  −0.434442453
normx1 (by hand)=  −0.434442453
0 to x2 (h4 computationally)=  7.8511341
0 to x2 (h4 by hand)=  7.85113412
x2 to x3 (h5 computationally)=  1.85852893
x2 to x3 (h5 by hand)=  1.85853013
x2 to x3 (h5 by handdddd)=  1.85852975
x3 to x1 (h6 computationally)=  5.116775
x3 to x1 (h6 by hand)=  5.11677644
K22  (computationally) = 0.289633874
K22  (by hand) = 0.289633879
K32  (computationally) = 0.269107222
K32  (by hand) = 0.269107226
K21  (computationally) = 0.309370222
K21  (by hand) =0.309370222
K31  (computationally) = 0.307270177
K31  (by hand) = 0.307270498
K33  (computationally) = 0.254547289
K33  (by hand) = 0.254548179
K11  (computationally) = 0.532197183
K11  (by hand) = 0.532199988
x2= 0.115657968
x3= 0.131786001
x1=  0.246324126
T by hand=1386.76494
V by hand=−270.028986
m2=48.72
R=  0.0462091341
Energywithoutmass = 1944.75825
Energy=  42 184.7583
```

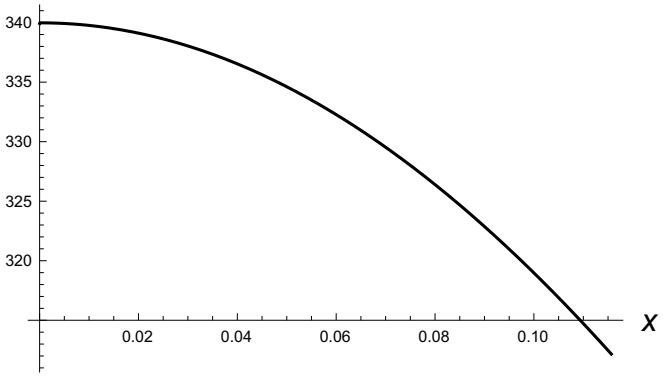
```
R=  0.0462091341
normx2 (computationally)=  -0.607761044
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161273
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442451
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.84965769
0 to x2 (h4 by hand)=  7.84965702
x2 to x3 (h5 computationally)=  1.8581763
x2 to x3 (h5 by hand)=  1.85817897
x2 to x3 (h5 by handdddd)=  1.85817711
x3 to x1 (h6 computationally)=  5.11577113
x3 to x1 (h6 by hand)=  5.1157723
K22  (computationally) = 0.289606457
K22  (by hand) = 0.289606272
K32  (computationally) = 0.269081864
K32  (by hand) = 0.269081737
K21  (computationally) = 0.309340543
K21  (by hand) =0.309340526
K31  (computationally) = 0.307241254
K31  (by hand) = 0.307241958
K33  (computationally) = 0.2545249
K33  (by hand) = 0.254526795
K11  (computationally) = 0.53214525
K11  (by hand) = 0.532147707
x2= 0.115668834
x3= 0.131798466
x1=  0.246348154
T by hand=1386.49621
V by hand=-270.003343
m2=48.711
R=  0.0462091341
Energywithoutmass = 1944.75751
Energy=  42184.7575
R=  0.0462091341
```

```
normx2 (computationally)=  -0.607761059
normx2 (by hand)=  -0.607761206
normx3 (computationally)=  -0.543161272
normx3 (by hand)=  -0.543161289
normx1 (computationally)=  -0.434442451
normx1 (by hand)=  -0.434442453
0 to x2 (h4 computationally)=  7.8498218
0 to x2 (h4 by hand)=  7.84982125
x2 to x3 (h5 computationally)=  1.8582154
x2 to x3 (h5 by hand)=  1.85821786
x2 to x3 (h5 by handdddd)=  1.85821622
x3 to x1 (h6 computationally)=  5.11588265
x3 to x1 (h6 by hand)=  5.11588384
K22  (computationally) = 0.289609523
K22  (by hand) = 0.289609371
K32  (computationally) = 0.269084693
K32  (by hand) = 0.269084588
K21  (computationally) = 0.309343841
K21  (by hand) =0.309343827
K31  (computationally) = 0.307244449
K31  (by hand) = 0.307245097
K33  (computationally) = 0.254527342
K33  (by hand) = 0.254529094
K11  (computationally) = 0.532150982
K11  (by hand) = 0.53215345
x2= 0.115667627
x3= 0.131797081
x1=  0.246345484
T by hand=1386.52606
V by hand=-270.006173
m2=48.712
R=  0.0462091341
Energywithoutmass = 1944.75759
Energy=  42 184.7576
x1=0.246345484
x3=0.131797081
```

```
x2=0.115667627
Kinetic Energy=1386.52606
Potential Energy=-270.006173
Volume Energy=828.23771
```

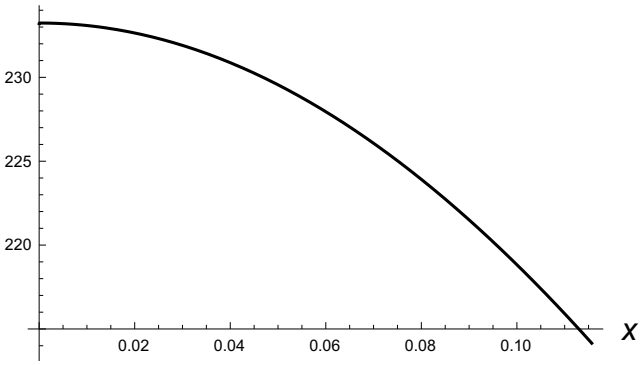
$(f_2(x)/x)^{3/2}$

Out[ ]=



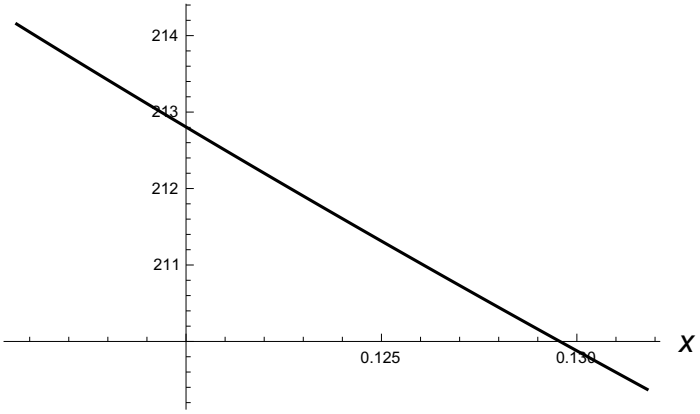
$(f_{2\,bar}(x)/x)^{3/2}$

Out[ ]=

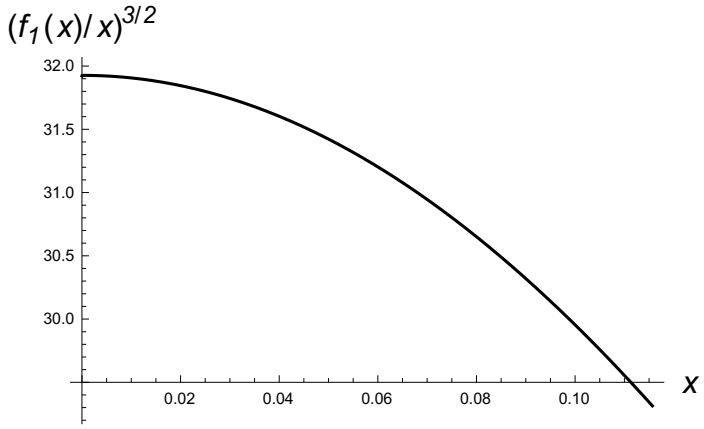


$(f_{2\,bar}(x)/x)^{3/2}$

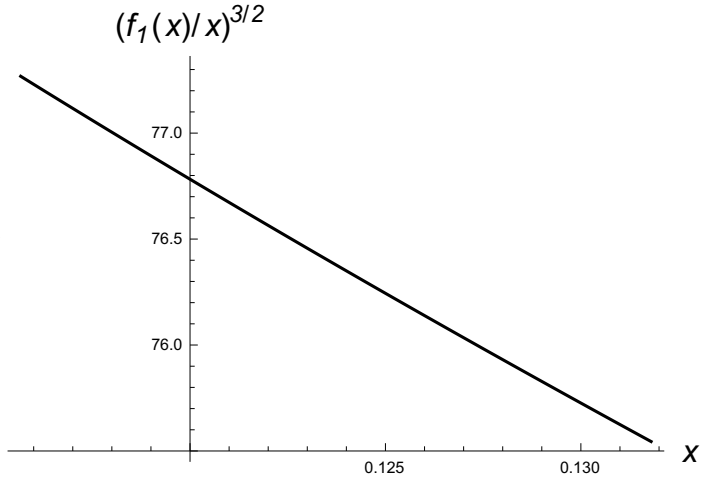
Out[ ]=



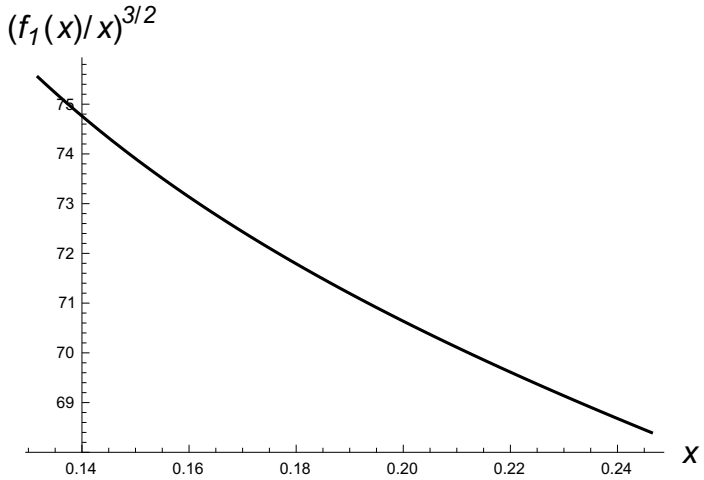
Out[ ]=


$(f_1(x)/x)^{3/2}$
32.0
31.5
31.0
30.5
30.0
0.02
0.04
0.06
0.08
0.10
x

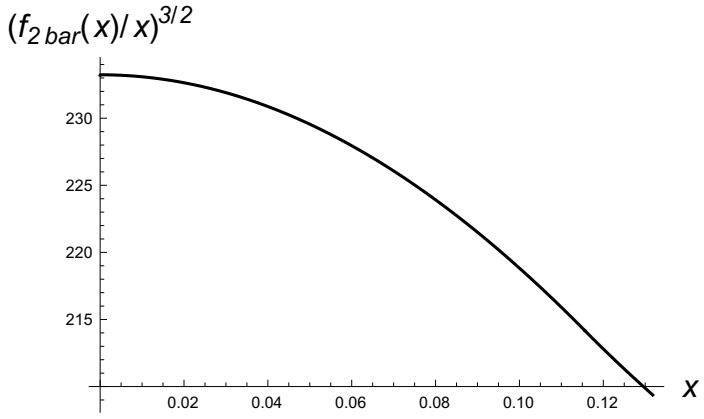
Out[ ]=


$(f_1(x)/x)^{3/2}$
77.0
76.5
76.0
0.125
0.130
x

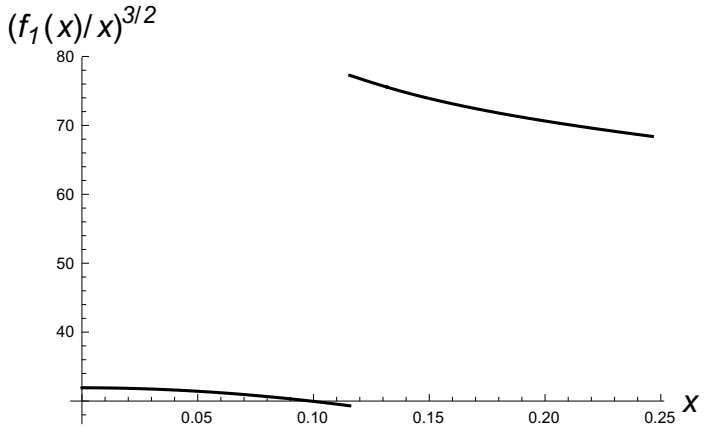
Out[ ]=


$(f_1(x)/x)^{3/2}$
75
74
73
72
71
70
69
0.14
0.16
0.18
0.20
0.22
0.24
x

Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
230
225
220
215
0.02
0.04
0.06
0.08
0.10
0.12
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
80
70
60
50
40
0.05
0.10
0.15
0.20
0.25
x